WEB PROGRAMMING

Project Proposal

Section Number	
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Project Title	Taskloom: A Lightweight Project Management App

Project Scope

Taskloom is a web-based project management tool, similar to a lighter version of Jira or Trello, built on the MERN stack (MongoDB, Express, React and Node.js). Teams create a workspace, split their work into projects, and track tasks on a Kanban board where cards are moved between columns by drag and drop.

Most small teams, including student groups, keep track of work in WhatsApp chats and shared documents. Tasks get lost and nobody is sure who owns what. Taskloom gives these teams one shared board with clear ownership, roles for each member, comments on every task and a full history of changes. Jira does this too, but it is heavy and takes time to learn. Taskloom keeps only the parts a small team actually uses.

The project covers the full stack: a REST API in Express, a MongoDB database with separate data per workspace, a React front end, authentication (including Google login), role-based access control and email notifications. The finished app will be deployed online.

Main Functionalities

- **User accounts:** Sign up and log in with email and password. Passwords are hashed and sessions use JWT tokens.
- **Google login:** Sign in with a Google account through OAuth 2.0.
- **Workspaces:** Each team gets its own workspace. Data from one workspace is never visible in another.
- **Projects:** A workspace holds multiple projects, and each project has its own board.
- **Tasks:** Tasks have a title, description, assignee, due date, priority and labels.
- **Kanban board:** Tasks sit in columns such as To Do, In Progress, Review and Done. Columns can be renamed or added.
- **Drag and drop:** Tasks can be dragged within a column or across columns to change their status.
- **Team invitations:** Members are invited by email through a link that expires after 48 hours.
- **Role-based permissions:** Four roles (Owner, Admin, Member, Viewer) decide who can view, edit, delete or manage members. Checks run on the server.
- **Comments:** Members can discuss a task in its comment section and @mention teammates.
- **Activity history:** Every change to a task is logged, so you can see who created, moved or edited it and when.
- **Email notifications:** Emails are sent for invitations, new assignments and mentions.
- **Search and filtering:** Search tasks by text and filter by assignee, label, priority or due date.
- **Dark mode:** Light and dark themes, saved per user.
- **Live updates (if time allows):** When one person moves a card, it moves on everyone else's screen too, using Socket.IO.